ActInf Textbook Group ~ Cohort 2 ~ Meeting 21 (Chapter 9, part 2)

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Hello, it's March 22nd, 2023. We're in Cohort 2 in our second discussion of Chapter 9.

We're going to jump right in to anything that people are thinking about related to Chapter 9 or just in general.

So whomever would like to bring something up, go for it.

Yeah. So I have a question that you said last time, I don't know whether it was in what context it exactly was,

that you would say, we talked about, or you talked about affordances, and you said that policies are not affordances.

And then I was wondering what then affordances are, because I understood policies are not affordances because they are in the future.

But then I was thinking what that should mean, because the present is a point in time, infinitely small,

which would imply that affordances doesn't exist. Or how would you see this?

So if one thinks that an affordance is, if we are in one state, and the policy is a sequence of states in the future,

but then even if we have this embedded states, I can be in the state of learning, for instance, active inference,

which also implies some actions in the future.

So I was a little bit confused about this notion, and why you would think that policies are not affordances.

Maybe I misunderstood.

Thanks.

Thanks.

Ali, want to give a first thought, and then I'll be happy to add something.

Well, yeah, actually, well, in fact, in active inference literature, there are some divergence between the way terms used in this context,

as compared to their other connotations or other meanings in some other context, such as reinforcement learning and so on.

So basically, in active inference, by policy, they generally mean a sequence of actions.

But for instance, in aria learning policy is defined as just a single action.

And affordance, again, is a kind of problematic term because we have different notions of affordance,

for example, in ecological psychology or in some other areas.

But to the best of my understanding, in active inference, affordance refers simply to a single action in some cases.

But in some other cases, it can take on some other additional layers of meaning.

So, for example, we can affordances or pragmatic affordances and they somehow conflated with each other and they can be confusing to delineate which notion of affordances is referred to here in this context.

So, yeah, those are pretty much the basic or at least some of the points that can help delineate those two concepts, affordance and policy.

But I'm sure Daniel would have some other thoughts on that as well.

Awesome. Yes. A lot of good points.

So, big picture, there are divergences between how the terms are used in ACTIMF and in reinforcement ecological psychology.

We hope and basically believe that there's coherence in how the terms are used, but that doesn't guarantee coherence within and across areas which themselves are not necessarily internally or externally consistent.

So, policies are sequences of actions over a given time horizon.

Affordances are the instantaneous single action possibilities described by the E variable.

So, map not territory.

If you're in a maze and you can be going up, down, left, right, then those are your affordances.

And over a time horizon of four, you could have policies of a time horizon of four with all the combinatorial possibilities of up, down, left, right, to the fourth power.

Then there are those sometimes informally applied statements like a pragmatic affordance or an epistemic affordance,

which is like when action is being taken that is oriented around epistemic value.

But this is kind of like a modifier, like it's a risky affordance or it's a, you know, fun affordance, not necessarily like a special subtype in a formal way.

And it's described by the E variable.

You could have a nested generative model.

So, like Livestream 42 with a slam, simultaneous localization and mapping.

At the lower level, the robot has up, down, left, right affordances.

At the higher level, the policy selection, the affordances have to do with locations moving amongst locations.

And so, you can have nested affordances because you can have nested models.

That's not any controversial or hot take on the system itself.

It's just describing the model.

And then what we can look into more and what Paolo is researching and some others is explicitly in contrast with the ecological psychology concept of the Gibsonian affordance use.

What was also called like affordance 1.0, 2.0.

And then Maxwell Ramstad taking the perspective that active inference affordances are affordances 3.0.

The Gibsonian advocates emphasize that affordances are directly perceived action capacities.

So, for them, indeed, it is not making sense that there could be affordances in the future.

But people have also considered planning from an ecological psychology perspective.

But their emphasis is on the direct perception of action possibilities.

Which, coming from the E-vector perspective, it's kind of like take it or leave it.

The E-vector might be known in a metacognitive way, might be directly perceived, or it might not.

The E-vector is with the capacities for action.

Here in step-by-step.

The E-vector is the habit.

These are the actions that can be taken, are enumerated, and the values in the cells reflect the prior on action.

We can see that in the textbook, in the figures for chapter 5.

So, here we have a dopamine-mediated, so neuromodulator-mediated balance between two modes of policy selection.

Here in some dopaminergic regions in the mammalian brain.

So, in both cases, observations are coming in.

In the indirect pathway, the E-vector is basically passed along.

And the policy posterior, which could be sampled from neutrally.

Or, you could have a shaky hand and erase differences in the policy posterior.

Or, you could have a precise hand and always select the maximum element, maximum posterior action probability.

In the indirect pathway, the habit vector is basically just passed along.

So, action selection resembles action prior.

Whereas, in the expected free energy deliberative type 2 archetype of action selection.

There's this sharpening or updating of the action prior into a policy posterior that's been re-weighted by expected free energy.

This is highly schematic, and not all the variables are shown in many cases.

It's just schema.

So, habits is something bottom-up.

Oh, sorry.

And the G vector is something top-down.

So, that's planned.

And the other one is routine or something.

Yes.

Habits and crumb.

Yes.

Yes.

Yes.

As always, these are aspects of generative models, not of organisms.

So, it's hard because we want to be referring to the habits and the deliberation patterns that cognitive systems do.

But also be aware that like E and G are phenomena of our model.

They're not phenomena of the world.

But that being said, the role of E is to, through its membership or composition, enumerate what instantaneous actions are possible.

If there's up, down, left, right, then the E vector is going to have four elements.

If there's nine directions it can move, then E has a length of nine.

The values within the E vector are the prior on those actions being selected in an instance.

So, if there was a uniform E vector across, up, down, left, right, 1, 1, 1, 1,

then there's no prior asymmetry towards taking any of those actions.

Whereas, if the E vector was 10, 1, 1, 1, 1, then that is equivalent to saying like, that in the absence of further expected free energy updates,

that action prior will be recapitulated in future action.

And so, you can imagine a model where E is updatable or not.

Like, a simple updating model for E would be every time you take an action, drop a pebble into that urn.

So, that's the learning as counting strategy, which works for many things.

So, that the action prior comes to embody previous action posteriors.

That's habit.

And so, what initially may have required a deliberative expected free energy expenditure

becomes canalized or entrenched in E,

allowing, for example, skill or skillful performance

to occur without the deliberative high dopamine setting in figure 5.4.

Here, the dopamine variable is gamma.

And, yes, two of the places to look for further discussion on that.

Some of it is the ontology project, where with Paolo and others,

we're talking about like, Ramstead and Camaro

and all of these different perspectives on affordance and representation.

And also, there's been an interesting discussion about like, similarities and differences

with the continuous and the discrete time model.

So, the affordance is 3.0 discussion.

And then the other angle for critical perspectives on the act-inf strategies

and counter rebuttals is the Markov blanket trick.

Ali?

Ali?

I just wanted to point to a kind of possible point of confusion

that can arise when dealing with these ABC, sorry, DE matrices and vectors,

because sometimes it's difficult to remember which matrix or which vector encodes what.

So, one simple mnemonic that I found helpful in remembering those notational conventions

is as we go from A to E, we move from instantaneous observations

into our prior and learned and habit beliefs about the world.

So, for instance, A encodes our beliefs about the relationship between the hidden states

and the observable outcomes.

B is about the likelihood of the transition.

C encodes our preferences.

D encodes our beliefs about initial hidden states.

And finally, E encodes our habits.

So, yeah, I think that might help to remember those terms.

Awesome.

And G, some kind of metaphysical expectations or something like this?
Or what is G then?
Well, it's all metaphysical.
They're all informational because they're all variables.
Although one can take the Chris Fields perspective, information as physics.
Expected free energy, right?
Yeah, yeah, yeah.
But that being said, we continue down the line with F as variational free energy
and G as expected free energy.
Yes.
However, these are all symbols.
And there's sort of a common set of symbols in some sense,
but they're by no means necessarily these letters.
It kind of goes without saying, but it definitely can be a point of confusion.
But let's look at our favorite step-by-step.
Pull in the image for the 100th time.
Then we have S and O.
O, observation.
S, hidden state.
D, prior on hidden states.
A, the emission recognition matrix between S and O.
This is the tale of two densities.
Because A can go from S to O, emitting in the generative direction,
plausible observations from hidden states.
Or A can be used in its recognition capacity to update S based upon incoming observations.
So, in the upper left static perception case, all you need to appeal to is A and D, S and O.
Then, dynamic perception.
It's still passive, but it's changing through time.
And what changes the hidden state through time is B, the transition matrix on hidden states.
To enable policy selection, at least two other things have to come into play.
Well, at least one, but really two.
The one that must come into play is simply the enumeration of action capacities,
which is listed in the membership of E,
and the cells contain quantitative information on the prior probability of those actions being selected,
a.k.a. habit.
You could have E and just be taking actions.
However, what is it that steers the ship on which actions are selected?
Well, the way we do that in active inference is not by proposing a utility function

and maximizing utility function and reward function,
but to encode through C our preferences, what we expect slash prefer.
We talk about them as preferences because they play a functional role teleologically as preferences
in the sense that pragmatic behavior reduces the divergence between observations and preferences.
So, in the sense that they are preferences, they are that.
But what C does is it's like a thumb on the scale of observations,
not by changing anything downstairs with a tail of two densities,
but rather through expected free energy,
selecting policies that are expected to reduce the divergence between observations and expectations.
And by way of reducing divergences between observations and expectations about observations,
we acquire pragmatic value oriented towards C,
in the extreme case, fully realizing C,
exactly having a homeostatic temperature,
exactly having how much money we want in the bank, whatever it is.
And so, functionally, C is called preferences.
And then there's this little extra piece on the right
that's also reflected in that dopaminergic image
with gamma as a temperature.
Gamma and beta as one over each other.
And those gamma being a parameterization of a big gamma distribution.
And them being one over each other.
So, whether you choose to think about it as like temperature or inverse temperature,
that is a precision on policy selection.
And then last thought on this,
this is not the only active inference model.
You could have a precision on anything.
You could have a precision on D.
You could have a precision on A.
You could have a precision on B.
So, these are like some birds I've seen.
This is not the overall exhaustive zoo of all possible birds or the one possible bird.
These are like certain configurations of base graphs,
which we can use a controlled ontology to describe
so that we can apply them in cybernetic contexts.
For example, designing ecosystems of shared intelligence.
Ali?
Ali?
I also believe beta is technically called stochasticity.

So, in some literature,
they explicitly use this term, stochasticity,
to refer to that beta parameter,
which sometimes, for the sake of simplicity,
is equated to one
in order to not complicate the distributions,
especially the Gaussian distributions.
But sometimes,
it can be an arbitrary value of stochasticity.
Yes.
Thank you.
And when we see the learning,
figure 7.10,
here,
learning in a Bayesian setting
can be seen as having a prior on that variable.
So,
if we have a fixed parameter,
D,
then it's just like a,
like a Dirac delta
hashtag category theory.
But it's like one point
that's just fixed
and it's a fixed value.
There's two states
and it's one,
zero.
So, just,
it is the first state.
Or,
you could imagine
the instantiation of that prior
parameterization
to be drawn
from another distribution.
and then,
you could imagine learning

or updating
within that setting.
But if you have just a fixed parameter,
then it,
it can't be updated.
but if you have a fixed parameter
or associated with a prior,
then you've opened the door to learning
and invoked all of these other
kind of Bayesian considerations,
like making sure that your prior
is not simply being recapitulated in the output.
So,
like testing for different priors
and doing all these different things
for standards,
rigorous Bayesian statistics.
Michael?
Yeah.
So,
in this graph
that you just showed,
the,
an affordance
would then
be the relation
between S and O?
No.
S and O,
hidden states
and observations.
So,
A is the tail of two densities.
Here's the temperature of the room,
S,
and the thermometer,
O.
So,

the recognition density
is what's the temperature
given the reading
and then the generative density
direction
is
from the temperature
what's the thermometer.
So,
the,
the downstairs
is all passive inference.
Downstairs,
passive inference.
here's our prior
on the temperature
in the room,
temperature in the room
changing through time.
We're not doing anything about it
and at every time step
it's emitting
an observation.
Where action comes in
is in the upstairs
with E,
the actions
that we can take
and associated
habituality
around them
and then
as to do more
than simply
recapitulate habit
we have preferences
which
have a semantics

of being
expectations
about
observations.
I expect
myself to be
homeostatic
and I'm going to
take actions
that make that
realized.
So,
reward learning
paradigm.
I'm rewarded
by having
homeostatic temperature
and I follow
policies
that are rewarding.
active inference
self-evidencing
paradigm.
I expect
to be
homeostatic
and so
I take
action selection
to reduce
divergences
between
the observations
that I do get
and what I
expect
slash prefer.
So,

how does
the
can the notion
of affordances
then be explained
with these
graphs that you
showed?
Is it a
vertical slide
through this
architecture
or
I mean
it has to do
with the O
and with
preferences
that's what
you just said,
right?
The actions
are taken
I mean
in a way
the actions
are taken
in the
adaptive case
in service
pragmatically
of reducing
the divergence
between
C and O.
That is
pragmatic value
is reducing

that and that's
with the KL
divergence
we're reducing
the divergence
between
C and O.
That's pragmatic
value.
Epistemic value
is reducing
our uncertainty
about
different things.
Where are the
affordances
can it be seen
as like a
slice in time
in this graph?
So,
the affordances
are enumerated
in the E
variable
and in
every single
time step
they are going
to be
crunched
by G
and then
B
let's just
say that
there are
two

affordances.
we can
turn the
heater
on
or we
can
have
the
heater
off.
B
is going
to have
two
slices.
B
is like
a tensor.
It can
be any
the dimensionality
is really
important and
looking at
the notebooks
and the
code around
these things
is some
of the
best way
to get
intuition
for this.
But
basically
B

has a
slice
for
every
affordance.
So there's
like a
B
like a
transition
matrix
for the
heater
being
off
and then
there's
a
transition
matrix
for the
heater
being
on.
So
like
let's
just
say
that
in
our
hidden
state
there's
three
temperatures
cold

medium
and
hot.
So
again
map
not
territory
we're
not
saying
that
there
isn't
a
quantitative
temperature
in
the
room
it
just
we're
choosing
to
model
the
hidden
state
with
three
discrete
states
and
so
if
the
heater

is
off
the
B
matrix
let's
just
say
looks
like
the
identity
matrix
so
if
the
heater
is
off
the
room's
temperature
does
not
change
if
the
heater
is
on
we
bump
from
cold
to
medium
medium
to

hot
and
we
stay
in
hot
and
so
then
the
ones
would
be
like
in
different
locations
on
that
slice
of
B
what do
you think
about
that
yeah
so
trying
to
digest
so
so
you're
saying
the
affordances
are

encoded
in the
transitions
between
states
the
affordances
are
most
plainly
listed
in
the
e
variable
however
every
affordance
is
also
represented
with
a
different
transitions
different
transition
slice
in the
world
if
there
was
just
one
transition
matrix
then

our
action
wouldn't
do
anything
because
the
world
is
just
going
to
change
the
way
it's
going
to
change
no
matter
what
we
select
so
then
all
policies
would
be
equally
meaningless
in
service
of
reducing
the
divergence

between
our
preferences
and
observations
so
again
not
a
feature
of
the
real
world
that
there's
actually
a
B
matrix
out
there
that
has
a
number
of
slices
equivalent
to
the
number
of
actions
that
can
be
taken

but
given
the
dimensionality
of
E
there
is a
slice
corresponding
to
each
element
of
E
in
B
and
then
if
affordance
1
is
taken
slice
1
is
used
in
the
transition
if
affordance
2
is
taken
slice
2

is
used
in
passing
S
forward
okay
and
what
what
if
there
is
somehow
an
implication
if
I'm
in
right
now
and
there's
an
implication
for
the
future
and
well
perhaps
not
really
a
deterministic
one
but
so

I
decide
with
my
decision
right
now
what
happens
in
the
future
but
the
future
would
not
be
in
affordance
in
that
sense
even
though
it's
part
of
maybe
a
policy
it's
not
an
affordance
because
it's
not

related
to
that
transition
that
is
immediately
activated
that's
right
now
the
case
something
like
this
that
that's
a
commendable
compatibilist
position
between
the
ecological
psychologist
and
active
inference
is
like
saying
affordances
are
real
in
the
moment

and
so
policies
which
also
include
actions
that
are
taken
potentially
in
the
future
like
imagined
actions
those
are
also
those
will
be
real
in
their
moment
and
they're
cognitively
real
in
this
moment
and
that's
why
this

also
invokes
the
discussion
around
representationalism
because
if one
takes a
realist
perspective
and says
well this
is what
they're
doing
they're
evaluating
all
counterfactuals
!
in the
organism's
brain
or in
the
computer
program
or
something
versus
is this
just an
instrumental
analytical
framework
that doesn't
actually

mean
that there's
an e-vector
or the
enumeration
of
counterfactual
policies
at all
so the
realist
angle
would be
this
is
the
causal
architecture
and
patterns
in this
base
graph
will be
associated
with
for example
neural
activity
patterns
which can
be understood
as neural
representations
of these
quantities
and
processes

the
instrumentalist
is under
absolutely
no
compulsion
for this
architecture
to reflect
anything
in specific
about
the slime
mold
the ant
colony
the
civilization
it's not
like there
actually has
to be
the
enumeration
of
counterfactuals
about
hypothetical
action
sequences
but we
list those
as a
convenience
and then
we can
apply
tree

surge
branching
time
active
inference
etc
etc
as
heuristics
in the
space
that is
combinatorially
constructed
but that
from an
instrumentalist
point of
view
doesn't
mean
that's
what the
system
is
doing
and
that is
like
at
one of
the
heart
of the
tension
because
the
Gibsonian

affordance
concept
centers
a
quintessentially
realist
phenomena
which is
what the
organism is
actually
instantaneously
perceiving
in terms
of action
capacities
chairs are
for sitting
which means
potentially
under like
some
people's
interpretation
of radical
and activism
like if
you don't
see the
chair
you're
not having
that
instantaneous
perception
of action
capacity
whereas

an active
inference
especially
from an
instrumental
reading
we're
basically
on the
plus side
completely
free
from that
but on
the downside
people need
to be very
careful
that they
don't
sneak
the
realism
back
in
and say
well
here's
going to
be the
central
nervous
system
just
conceptually
decision
making
and here's

going to
be the
nested
model
of the
hand
and then
it's all
good
they go
off on
their
instrumental
way
and then
all of a
sudden
they're
interpreting
this as
the
elbow
as if
this
architecture
was
reflecting
for
example
something
anatomical
so
instrumentalism
allows one
to fly
freely
with this
analytical

framework
and leave
a lot
of baggage
at the
door
however
in the
case where
realist
conclusions
are being
sought
to be
constructed
one has
to be
careful
that they
weren't
actually
taking a
covert
realism
along
for the
ride
the
good
news
is I
think
people
can
do
that
because
they

don't
have
any
problems
with
linear
regression
but
then again
some
do
in
some
cases
but
the
architecture
of that
Bayes
graph
can be
seen of
similar
to
structural
equation
modeling
it's not
exactly
the same
but
having
edges
and nodes
is not
the
system's
mechanics

but
given
this
statistical
artifact
it's
very
easy
to
say
that
this
causes
that
instead
of
being
proper
and
saying
it's
associated
with
that
especially
because
you get
into
directed
edges
it's
like
well
this
is
unidirectionally
associated
with that

through
time
it's
like
come
on
then
just
say
it
what
do
you
think
yeah
if
you
ask
me
I'm
I'm
digesting
it
I need
some
more
time
to
get
all
this
a little
bit
clear
but
thanks
so
much

to
elaborate
on this
question
thank you
so much
yeah
this is
like
the
heart
of
the
textbook
and
the
topic
and
this
is
a
model
architecture
it's
not
the
only
model
architecture
but
being
able
to
interpret
what
these
model
architectures

are

helps

us

understand

a

lot

about

active

inference

and

then

I

guess

as

we

are

seeing

it

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shines

a lot

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interesting

light

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broader

questions

about

scientific

modeling

and

bigger

discussions

which

I mean

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return

to
chapter
nine
that's
the
metabasean
perspective
that's
why
we're
centering
this
ethological
moment
with
the
modeler's
construction
of
a
cognitive
model
whose
aboutness
is the
setup
the
experimenter
has
constructed
and
again
that
to
make
that
explicit
and

encoded
in our
model
structurally
is the
innovation
because
upon
inquiry
I don't
think this
is a
contentious
layout
like
if
inside
of the
dashed
line
there
was a
linear
regression
or an
SPM
model
and we
said
okay
we
gave
we
did
the
teammates
and we
had

the
rat
hooked
up
to
the
whatever
optogenetic
mix
or
to
the
confocal
microscope
or
to
the
EEG
or
to
the
video
camera
or
to
the
ultrasound
it's
like
you
got
all
these
scientific
tools
okay
we
set

up
an
experimental
context
and
then
we
observe
behavior
now
what
is
it
that
was
in
between
the
stimuli
and
the
behavior
well
obviously
it
was
the
mouse
or
the
rat
itself
okay
so
we
want
to
have

at
the
very
least
some
kind
of
model
of
our
rat
that
is
able
to
perceive
experimental
stimuli
and
then
output
behavior
so
we
could
put
a
linear
regression
here
here's
temperature
on the
x-axis
and
likelihood
of
eating

cheese
on the
y-axis
and
we
make
a
regression
or
we
do
model
comparison
we
say
okay
is it
better
compatible
with a
linear
regression
or
a
breakpoint
regression
or a
quadratic
regression
and so
you have
the modeling
discussion
and they
go okay
we use
the BIC
and the

optimal
model
to describe
the relationship
between temperature
and cheese
eating
is a
breakpoint
regression
or it's just
a simple
linear regression
regression
would

someone
turn
around
and say
why
would
you
say
mice
are
linear
regressions
the
obvious
response
would be
no we
just
use
the
linear
regression

instrumentally
to describe
our data
check out
the method
section
look
again
we
followed
pretty
standard
process
of
modeling
so that
we were
able to
explain
data
but not
overfit
accuracy
minus
complexity
and
we
use
standard
software
tools
and
packages
so that
everybody
could replicate
it
at its

open
source
we're
definitely
not saying
that
mice
are
linear
regressions
but when
figure 4.3
goes into
the box
different
questions
arise
and
I'm not
even saying
that what
I have
just
proposed
is
infallible
I
think
people
can
have
a
different
perspective
on
that
for
example

they
could
say
well
yeah
if
you
do
a
linear
regression
you're
not
saying
the
mouse
is
a
linear
aggression
but
there's
something
different
about
active
inference
modeling
where
when
you
put
figure
4.3
inside
the
box
you've

taken
on
a
different
set
of
commitments
that's
a
totally
valid
point
to
make
if
somebody
can
point
to
those
commitments
and
but
simply
considering
active
inference
as like
a
software
package
or
an
analytics
framework
for
behavior
just

replace
figure
4.3
in your
mind's
eye
with
linear
aggression
and
ask
what
claims
are
valid
or
not
and
there
are
differences
between
the
models
so
there
may
be
certain
things
that
go
one
way
for
linear
aggressions
and

another
way
for
POMDP
however
there's
also
a
broad
set
of
criticisms
that
people
level
at
figure
4.3
that
are
throwing
out
the
linear
aggression
as
well
but
it's
a
format
that
people
are
more
familiar
with
I

assume
this
this
explains
much
more
potentially
at
least
you
could
generate
a
POMDP
with
zero
explanatory
predictive
!
value
and
you
could
have
a
linear
regression
with
a
good
R
squared
or
not
it's
not
a
function

of
the
type
of
model
selected
in
fact
the
space
of
POMDP
is
just
from
a
parametric
perspective
it's
much
larger
like
here
this
is
still
a
relatively
simple
this
is
even
without
showing
the
dimensionality
of
these

variables
like
if
there's
10
affordances
B
is
going
to
have
10
slices
and
then
if
S
has
10
hidden
even
as
it's
10
discrete
options
so
you're
talking
about
some
large
tensors
and
parameterizing
them
from
data

is
non
trivial
that's
what
chapter
nine
is
about
and
that's
why
we
have
often
called
for
the
development
of
statistical
power
analysis
techniques
because
even
simple
models
have
many
free
parameters
that
might
be
related
in
complex

ways
like
if
C
is
like
within
this
range
then
changes
in
A
do
or
don't
do
that
but
if
A
becomes
more
ambiguous
then
this
consequence
happens
over
here
and
those
kinds
of
rippling
effects
in
the

model
are
not
constrained
to
which
edges
are
drawn
so
we
need
much
better
statistical
power
analysis
techniques
in
order
to
anticipate
the
kinds
of
statistical
considerations
that
people
use
for
linear
regressions
and
RNA-seq
analyses
every
day

but
there's
nothing
about
this
model
that
makes
it
fit
any
given
data
set
better
or
worse
these
are
all
modelers
choice
any
closing
thoughts
or
questions
on
nine
before
we
scan
10
I
also
think
it'll
be

interesting
to
see
like
as
we
have
more
notebooks
and
accessible
software
packages
that
will
help
support
this
deflationary
perspective
on
active
inference
because
it'll
be like
well
what
part
of
this
tiger
is
the
B
matrix
and
we'll

have
like
a
notebook
it's
like
well
here's
the
B
matrix
in
the
notebook
but
it's
not
in
the
world
and
then
just
coming
at
that
nexus
many
many
times
will
be
useful
but
also
I
think
including

people
in
empirical
research
is
going
to
help
with
that
because
for
one
who
hasn't
read
a
paper
and
or
for
one
who
hasn't
participated
in
empirical
behavioral
research
neither
of which
are
bad
or
good
states
just
saying

some
people
have
these
experiences
and
some
don't
that
separation
of
statistical
models
from
reality
like
mice
are
not
linear
aggressions
mice
are
not
POMDPs
that
is
definitely
like
a
posture
and
I
don't
mean
it's
a
facade

it's
a
stance
that
is
built
through
statistical
investigation
at the
very least
reading it
if not
actually
doing it
oneself
like
once
one
sees
that
there's
a
hundred
or
an
open
ended
number
of
ways
to
model
just
the
thermometer
and
the

true
temperature
of
the
room
that
person
will
think
differently
about
map
and
territory
this
is the
final
chapter
10
in
general
we
are
least
aware
of
what
our
minds
do
best
by
Dr.
Minsky
chapter
10
we're
going to

wrap up
active
inference
main
theoretical
points
from the
first
half
of the
book
chapters
one
through
five
and
the
practical
implementations
from the
second
part
six
through
ten
then
we
connect
the
dots
abstracting
away
from
specific
models
discussed
to
focus

on
integrative
aspects
a
benefit
of
active
is it
provides
a
complete
solution
to
the
adaptive
problems
that
sentient
organisms
have
to
solve
this
is
definitely
a
sentence
that
we've
discussed
in
prior
cohorts
does
it
provide
a
complete

solution
or
does
it
set
the
table
for
a
useful
solution
or
some
other
way
of
saying
that
you
know
have
the
previous
nine
chapters
literally
resolved
all
the
problems
or
have
they
helped
us
learn
the
linguistics

and
techniques
that
help
us
frame
the
problems
so
that
we
can
come
to
heuristic
solutions
but
again
these
are
the
kinds
of
things
we
can
discuss
10.2
wrapping
up
this
par
at all
textbook
offers
a
systematic
account

of
the
theoretical
underpinnings
part
one
practical
implementations
of
active
inference
part
two
they
review
chapter
one
chapter
two
long
low
road
chapter
three
high
road
well
do you
need
your
dinner
tomorrow
chapter
five
highlights
some of
the
applications

and
research
in
active
inference
in
the
mammalian

nervous
system
chapter
six
the
recipe
to
design
active
inference
models
chapter
seven
and
eight
discrete
and
continuous
time
generative
models
in
active
inference
chapter
nine
active
inference
in

context
of
model
based
data
analysis
connecting
the
dots
integrative
perspectives
on
active
inference
Daniel
Dennett
and
the
whole
iguana
!
Hey
instead
of
going
deep
down
the
rabbit
hole
with
the
prospective
working
memory
module
subsystem
and

then
developing
this
whole
just
so
story
backwards
from
that
how
about
modeling
a
complete
cognitive
creature
and
environmental
niche
for it
to
cope
with
they
talk
about
that
whole
iguana
perspective
and
the
way
that
diverse
cognitive
phenomena

attention
memory
anticipation
etc
etc
can
be
understood
of
using
this
active
inference
first
principles
framework
active
inference
doesn't
start
by
assembling
separate
predefined
cognitive
functions
perception
decision
making
and
planning
rather
it
starts
by
providing
a
complete

solution
or
approach
to
derive
implications
about
cognitive
functions
that
helps
us
identify
patterns
of
cognitive
phenomena
for
example
optimal
foraging
in
different
settings
finally
active
inference
offers
a
principled
means
of
understanding
corresponding
neural
computations
it
speaks

to
multiple
levels
of
mars
hierarchy
mars
levels
and
very
briefly
we'll
discuss
more
in the
coming
weeks
implications
for
psychological
function
as if
we were
sketching
or prompting
a
psychology
textbook
predictive
brains
minds
and
processing
what does
active
inference
have to
do

and
how
does
it
build
within
the
lineage
of
research
on
the
predictive
brain
predictive
mind
perception
and
different
psychological
and
philosophical
lineages
for example
going back
to
Helmholtz
and before
on
perception
Bayesian
brain
action
idea
motor
theory
cybernetics
optimal

control
theory
utility
in
decision
making
Bayesian
decision
theory
reinforcement
learning
planning
as
inference
behavior
and
bounded
rationality
free
energy
theory
of
bounded
rationality
valence
emotion
and
motivation
homeostasis
!
allostasis
interoceptive
processing
attention
salience
and epistemic
dynamics
rule

learning
causal
inference
and
fast
generalization
active
inference
and other
fields
open
directions
social
cultural
dynamics
machine
learning
and
robotics
and
summary
with
the
lord
of
the
rings
quote
ultimately
we are
confident
that you
will
continue
to
pursue
active
inference

in
some
form
okay
thanks
for
the chapter
nine
discussion
we will
come back
for the
coming two
weeks
in chapter
ten
and then
one final
session
so
thank you
all
five